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Scalar-on-distribution regression for assessing the impact of climate change on rice yield in Vietnam

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Abstract

In econometrics, the impact of climate change on agricultural yield has often been modeled using linear functional regression, where crop yield, a scalar response, is regressed on the temperature distribution over a given time period, treated as an ordinary functional parameter, along with other covariates. We explore alternative models that respect the distributional nature of the temperature distribution parameter. Replacing functional observations with the corresponding distributional ones is appropriate for phenomena that are insensitive to the temporal order of events. Since classical addition and scalar multiplication are unsuitable for density functions, alternative operations, spaces and corresponding regression techniques are required. Moreover, compositional data analysis suggests that such covariates should undergo appropriate log-ratio transformations before inclusion in the model. We compare a discrete approach, where temperature histograms are treated as compositional vectors, with a functional scalar-on-density regression using a Bayes space representation of temperature densities. We evaluate the strengths of each method in modeling rice yield in Vietnam, using data on daily temperature extremes. Additionally, we propose modeling climate change scenarios with perturbations of the initial density along a change direction curve informed by IPCC scenarios. The resulting rice yield marginal impact is then

quantified using a simple inner product between the density covariate parameter and the change direction curve. Our results indicate that while both approaches yield coherent findings, the scalar-on-density model outperforms the scalar-on-composition with an enhanced ability to accurately gauge the phenomenon’s scale.

Keywords: Compositional scalar-on-density regression, scalar-on-composition regression, Bayes spaces, compositional splines, functional regression, climate change, rice yield, Vietnam.

1 Introduction

Evaluating the impact of climate change on diverse socio-economic phenomena has become an increasingly compelling issue. To this end, temperature data is summarized and incorporated in various ways as a covariate in regression models. The typical raw data is made of daily minimum, maximum and means. Most models use simple temperature time averages (monthly, yearly, depending on applications). More generally, as the complexity of recorded data continues to grow, contemporary models increasingly involve intricate data objects. For example, another option is to view the curve of temperature evolving over time as a functional covariate as in [1]. Within the field of econometrics, some models employ temperature histograms or densities that capture the distribution over time periods, see for example [2] for a comprehensive review in the context of climate change econometrics.

Summarizing the temperature data by their distribution over a time-period induces a loss of information compared to using the full temporal sequence, which captures the order of temperature changes over time. Yet, for some agronomic applications, [3] argue that this piece of information is not crucial. They propose a functional linear model with temperature density as covariate and, in a second phase, approximate the densities by histograms. When using histograms as covariate, the sample space can be described by the set of vectors of relative bin frequencies (percentages), with positive components that sum to 100. When interpreting the percentages as the corresponding bin probabilities, the constant becomes 1, and the histogram represents a discrete distribution.

However the Schlenker and Roberts’ ([3]) linear functional regression model does not adequately account for the distributional nature of the functional density parameter. When the density is represented by a vector of D relative bin frequencies, their model applies the classical operations of $\mathbb{R}^D$ and the corresponding standard notion of linearity. Yet $\mathbb{R}^D$ is not a natural space for these vectors due to their inherent constraints: the components are positive and sum to 1. They are known as compositional vectors and naturally reside in a simplex rather than in $\mathbb{R}^D$. For example the classical addition applied to two relative bin frequencies vectors yields a vector which is not a relative bin frequency vector. A suitable linear structure for this space is provided by the Aitchison geometry, see [4] for an introduction. Compositional data analysis adapts

the concept of linearity enabling the definition of linear regression models that incorporate discrete distributions as covariates, yielding the scalar-on-composition regression model. These models transform distributional data into vectors of log-ratio of frequencies ensuring scale invariance. Scalar-on-composition regression models using the simplex representation are described for example in [5]. Finally another disadvantage of the model in [3] is the fact that they deal with the sum constraint by removing one bin in the model equation resulting in a very particular interpretation of the parameters in reference to that bin.

Moreover, assimilating the histogram data to a discrete density results in a loss of information about the order of the temperature bins since compositional data treatment is permutation invariant. For this reason, we propose to use rather a functional approach adapted to density functions. Random distributions are commonly reported as histogram-valued data or as samples. [6] propose a regression model involving histogram-valued covariates representing them by quantile functions. When random distributions are observed through their samples, a preprocessing step transforming these samples into density functions, allows for the application of functional data techniques. For this preprocessing, [7] propose a specialized class of splines, known as compositional splines or CB-splines, specifically designed to accommodate density constraints. [8] conduct a comprehensive review of methodologies for constructing regression models involving samples of probability density functions from a functional perspective. In functional data analysis, densities stand out as unique entities because of the constraints they must satisfy. For one-dimensional densities, the sample space can be defined as the space $\mathcal{D}$ of functions with positive values and a unit integral. Among the two main approaches discussed in [8], one focuses on representing densities within Bayes spaces $\mathcal{B}^2(a, b)$. Because densities in Bayes spaces serve as a continuous analogue of compositional vectors in the simplex, it is natural here to explore their potential, rather than alternative representation spaces such as the space of log-quantile densities or the object-oriented approaches, see [8]. Comparing the Bayes spaces functional approach with the discrete compositional method using histogram frequency vectors offers valuable insights into their respective advantages.

Bayes spaces, initially introduced by [9], later extended in [10], endow the space $\mathcal{D}$ of densities with a finite support $[a, b]$ with a Hilbert space structure. This space and structure can be viewed as a continuous version of the simplex and its associated operations. As for the log-ratio transformation, the functional centered log-ratio serves as the functional counterpart of the classical centered log-ratio transformation for vectors of a simplex. This concept is used for example in [11] to construct functional scalar-on-density regression models. Scalar-on-composition and scalar-on-density regression models can be viewed respectively as a discrete and smooth versions of the scalar-on-distribution regression model. Nevertheless the functional (smooth) approach implementation is more complex raising the natural question of whether the functional model offers a meaningful advantage over the discrete approach based on histogram data. To the best of our knowledge, no study has directly compared these two models.

While our analysis centers on modeling the impact of temperature data on rice yield in Vietnam, the proposed models have broader applicability across various application

domains, notably those related to climate change impact evaluation. For this reason, we develop tools within these models to assess the marginal impact of climate change on yield and its variability.

We do not present this application as a case study. Indeed, it does not account for regional variations in rice-growing seasons, the differing effects of daytime and nighttime maximum temperatures which tend to act in a reverse way, the distinction between wet and dry seasons, or the decline in land under rice cultivation. Instead, the aim is to highlight two alternative modeling approaches that differ from those commonly used in the climate change literature, as most studies rely solely on monthly averages. Producing reliable predictions would require a multidisciplinary team with access to detailed data.

The paper is organized as follows. Section 2 reviews the methodological tools involved in these discrete (scalar-on-composition) and smooth (scalar-on-density) regression models (in particular, simplex space and Bayes space structures, centered log-ratio transformations), as well as the construction of the compositional splines. Section 3 presents the rice yield and the weather data and explores their main features. Section 4 presents the scalar-on-composition and scalar-on-density regression models and their estimation results. It also provides an interpretation of the discrete and smooth parameters associated to the temperature distribution parameters. Section 5 presents our proposal for the construction of a climate change scenario and the computations of the mean and variance of its marginal impact. Based on the model fit, we perform the computations for the RCP2.6 scenario provided by IPPC for the end of the century. An illustration of these impacts on the dataset allows to reveal the interest of the functional approach. Section 6 then concludes.

2 Methodological reminders

The dataset central to our problem comprises observations of maximum and minimum daily temperatures spanning a 30-year period, from 1987 to 2016, across 63 provinces in Vietnam, summarized as yearly distributions. These temperature density distributions serve as key covariates within our regression model, aimed at identifying factors influencing rice yield in Vietnam over this timeframe. In the discrete approach, temperature covariates are represented as compositional vectors. We provide an overview of fundamental techniques for working with compositional vectors in Section 2.1. In the smooth approach, temperature covariates are represented as density functions and we review the construction of the Bayes space in Section 2.2. For the regression analysis, the discrete approach employs scalar-on-composition regression techniques, as presented by [5]. In contrast, the regression part of the functional approach necessitates a preliminary step to transform the temperature histograms into $\mathcal{B}^2$ elements using CB-splines smoothing. We briefly review CB-splines in Section 2.3 and CB-splines smoothing in Section 2.4.

2.1 Discrete densities as compositional vectors

Let us first recall that compositional data (hereafter referred to as CoDa) vectors can be defined as vectors consisting of D positive components that sum to one, forming elements of a simplex denoted by $\mathcal{S}^D$. A discrete density function associated to a random variable with a finite number of outcomes is typically represented by its probability mass function, or equivalently by the vector of probabilities of each of these outcomes which satisfies the same constraints as a CoDa vector. In Section 4.1, the histogram data will be assimilated to a discrete density.

The space $\mathcal{S}^D$ can be equipped with a vector space structure using the following operations, see e.g. [12]. For all $\mathbf{u}, \mathbf{v} \in \mathcal{S}^D$ and $\lambda \in \mathbb{R}$, the perturbation, respectively power multiplication, are defined by $\mathbf{u} \oplus \mathbf{v} = \mathcal{C}(u_1 v_1, \dots, u_D v_D)$, respectively by $\lambda \odot \mathbf{u} = \mathcal{C}(u_1^\lambda, \dots, u_D^\lambda)$, where $\mathcal{C}$ denotes the closure of a vector (division by the sum of its components).

The above operations enable the definition of a meaningful average of a sample of n compositional vectors $\mathbf{u}_i$ (for $i = 1$ to n) by $\bar{\mathbf{u}} = \frac{1}{n} \odot (\mathbf{u}_1 \oplus \dots \oplus \mathbf{u}_n)$ (thus the components of this average are the geometric averages of the corresponding sample's components).

The clr transformation of a vector $\mathbf{u} \in \mathcal{S}^D$ is defined by $\text{clr}(\mathbf{u}) = \mathbf{G}_D \ln \mathbf{u}$, where $\mathbf{G}_D = \mathbf{I}_D - \frac{1}{D} \mathbf{1}_D \mathbf{1}_D^T$, $\mathbf{I}_D$ is a $D \times D$ identity matrix, $\mathbf{1}_D$ is the D -vector of ones and where the logarithm of $\mathbf{u} \in \mathcal{S}^D$ is understood componentwise. For a vector $\mathbf{u}^*$ in the orthogonal space $\mathbf{1}_D^\perp$ (orthogonality with respect to the standard inner product of $\mathbb{R}^D$), the inverse clr transformation is defined by $\text{clr}^{-1}(\mathbf{u}^*) = \mathcal{C}(\exp(\mathbf{u}^*))$.

The simplex $\mathcal{S}^D$ of dimension $D - 1$ can be equipped with the Aitchison inner product $\langle \mathbf{u}, \mathbf{v} \rangle_A = \langle \text{clr}(\mathbf{u}), \text{clr}(\mathbf{v}) \rangle$, where the right hand side inner product is the standard inner product in $\mathbb{R}^D$.

2.2 Continuous densities as elements of the Bayes space

As outlined in [13], density functions supported in a bounded interval $[a, b]$ can be regarded as elements of the so-called Bayes space denoted by $\mathcal{B}^2([a, b])$ and comprising positive functions integrating to one on $[a, b]$ whose log-transform is square integrable. This concept corresponds to a particular case of that introduced in [10] for the reference measure being the Lebesgue measure.

This space can first be equipped with a vector space structure using the following operations. For any positive function f on $[a, b]$, let the closure $\mathcal{C}(f)$ of f be the unique density proportional to f . Subsequently, for any two functions f and g in $\mathcal{B}^2([a, b])$ and any real α , the addition and scalar multiplication operations are redefined as

- perturbation as $(f \oplus g)(t) = \mathcal{C}(f(t)g(t))$,
- powering as $(\alpha \odot f)(t) = \mathcal{C}(f(t)^\alpha)$.

The centered log-ratio (clr) transformation is defined for $f \in \mathcal{B}^2([a, b])$ and t in $[a, b]$ by

$$\text{clr} f(t) = \log f(t) - \frac{1}{b-a} \int_a^b \log f(u) du. \quad (1)$$

Through its construction, the clr transformation maps $\mathcal{B}^2([a, b])$ into the space $L_0^2([a, b])$ of square integrable functions on $[a, b]$ with a zero integral. The inverse transformation is well defined and can be expressed as follows, for a function $f_0 \in L_0^2([a, b])$,

$$\text{clr}^{-1}(f_0)(t) = \mathcal{C} \exp(f_0(t)).$$

$\mathcal{B}^2([a, b])$ can then be equipped with an inner product rendering the clr transformation isometric, for a corresponding choice of inner product in $L_0^2([a, b])$. We adopt the definition in [13] for the $\mathcal{B}^2([a, b])$ inner product, which differs by a constant from the inner product introduced in [10]:

$$\langle f, g \rangle_{\mathcal{B}^2(a, b)} = \int_a^b \text{clr} f(t) \text{clrg}(t) dt = \langle \text{clr} f, \text{clrg} \rangle_{L_0^2([a, b])}. \quad (2)$$

2.3 Reminder on spline basis for density representation

Spline functions are constructed by piecing together segments of polynomials of a specified degree connecting at specified knot points while adhering to prescribed smoothness conditions [see e.g. 14]. The knots make spline functions more flexible than polynomials, allowing a larger number of parameters without increasing the degree. This prevents excessive boundary oscillations. In approximation theory and functional data analysis, it is common to use B-splines for writing expansions of spline functions in a finite dimensional space [see e.g. 15]. While there exist other basis for this purpose such as the truncated polynomial basis which are easier to express, B-spline basis functions are appreciated for their local support property. In particular, this property results in a band matrix structure of the matrix to be inverted in a least squares approximation problem, which numerically improves computations.

In our context, aimed at approximating density functions, we require a specific type of constrained splines. One approach to constructing them is described in [13] using the so-called ZB-splines in $L_0^2([a, b])$ and corresponding CB-splines in $\mathcal{B}^2([a, b])$. As is common in many CoDa techniques, the procedure is based on a log-ratio transformation, specifically here, the clr introduced in Section 2.2. The process starts by constructing a basis of spline functions that fulfill the integral constraint within $L_0^2([a, b])$. These basis functions are then pulled back to $\mathcal{B}^2([a, b])$ by the inverse clr transformation.

For a given order d and a knot sequence $\Lambda = \{(\lambda_1, \dots, \lambda_g) : a < \lambda_1 < \dots < \lambda_g < b\}$ whose elements are called inside-knots, let S_d^Λ be the subspace of $L^2([a, b])$ of polynomial splines of order d (degree $k = d - 1$) and inside-knots $(\lambda_1, \dots, \lambda_g)$, see [15] for a complete description. S_d^Λ has dimension $d + g$ and its most popular basis is given by the set of so-called (normalized) B-splines functions which have a small support and good computational properties. For technical reasons, additional knots are introduced at the boundary: if k is the degree of the polynomial pieces ($d = k + 1$ the corresponding order), k knots equal to a are added at the beginning of the interval and k knots equal to b at the end. [13] construct a basis of so-called ZB-splines for the subspace $\mathcal{Z}_d^\Lambda = S_d^\Lambda \cap L_0^2([a, b])$ of dimension $g + k$, the loss of one dimension being due to the zero-integral constraint. The inverse clr of the ZB-basis functions are called the CB-basis functions. For this application, we use exclusively cubic splines for which $k = 3$ and $d = 4$. Equation (17) in [13] establishes the correspondence between the

representation of any function in $\mathcal{Z}_d^\Lambda$ within both basis systems. This correspondence proves invaluable as it facilitates the manipulation of ZB-splines using conventional code originally designed for B-splines.

2.4 Smoothing histograms with CB-splines

Our original temperature data comprises sequences of daily maximum and minimum temperatures for each province and year, measured in degrees Celsius. In order to apply the same technique as [13], we first preprocess the data into intermediate histogram representations subsequently transformed into smooth density functions using CB-splines as in [7]. For maximum temperatures, the data are grouped into 28 bins of length 1 within the interval $[a, b] = [12, 40]$. For minimum temperatures, the data are grouped into 22 bins of length 1 within the interval $[a, b] = [7, 29]$. The choice of 1 degree for the bin length is common in temperature studies and results in a reasonable simplex dimension for the histogram space.

The CB-spline smoothing step involves choosing a ZB-spline basis in L_0^2 and viewing the estimation of the clr transformed densities expressed in the ZB-basis as a penalized least squares regression. This regression smoothes the scatterplot of the clr transformed histogram frequencies as a function of the ZB-spline basis functions evaluated at the midpoints of the histograms bins. We employ cubic splines (in L_0^2) positioning the knots relative to the data points using quantiles as argued in [13].

To illustrate the basis and the effect of the number of knots, Figure 1 represents two sets of basis functions obtained respectively for 7 and 9 knots on the interval $[12, 40]$ positioned at the quantiles of the full set of daily maximum temperatures. The curves in the right panel represent the ZB basis in $L_0^2([a, b])$ and the left panel are the corresponding CB spline basis functions in $\mathcal{B}^2([a, b])$. The dimension of the ZB-spline basis is $7 + 3 = 10$ when using 7 inside knots (respectively, $9 + 3 = 12$ for 9 inside knots). The vertical dotted lines on the plots indicate the knots positions. We observe in the lower plots that moving from 7 to 9 knots results in an increased number of basis functions that concentrate around the mode of the distribution. This enhancement enables a more precise approximation of the densities, particularly in regions where our dataset features a higher density of temperature data points.

To ensure the existence and uniqueness of the least squares problem (full column rank of the collocation matrix), we enforce an upper limit on the number of knots. This upper bound is dictated by the two Schoenberg-Whitney conditions (see [16]). In our application, the first condition, both for maximum and minimum temperature, stipulates in particular that the number of knots must be less than or equal to the number of bins minus 3 (degree of splines). Moreover, it turns out that a number of knots higher than 11 results in the violation of the second Schoenberg-Whitney condition. We therefore select 9 knots in the sequel.

Smoothing with ZB-splines does not accommodate bins with zero counts because of the log transformation. To address this limitation, we implement a simple zero-replacement procedure: any zero count is substituted by 10^{-7} after which we apply the closure operator. The sensitivity to the replacement value has been checked with 10^{-8} and 10^{-5} turns out to be negligible. For the smoothing parameter selection, we opt for a generalized cross-validation using a regular grid of 100 points on a log-scale.

Fig. 1 CB-splines (left) and ZB-splines (right) with 7 inside knots (top) and 9 inside knots (bottom)

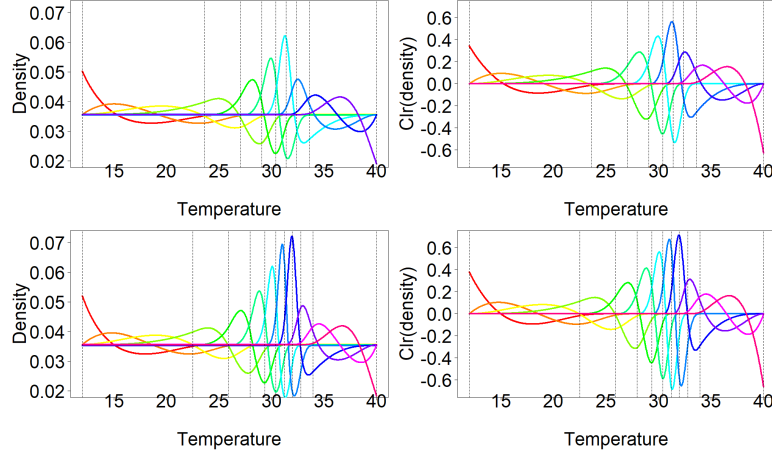
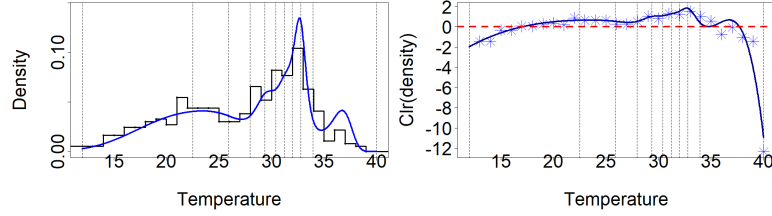


Fig. 2 Density of daily maximum temperature in 1995 in Yen Bai province (left) and its clr transform (right)



As an illustrative example, Figure 2 displays the histogram of the daily maximum temperatures in 1995 in the Yen Bai province (North-East of Vietnam), as well as the corresponding smooth density obtained by the above procedure on the left plot, and the smoothed clr transform on the right plot.

3 Data and exploratory analysis

3.1 Rice yield data

The dataset concerning rice yield is sourced from the International Rice Research Institute¹. The data set contains comprehensive information on annual rice production, harvested area, and rice yield at provincial level from 1987 to 2016. Rice yield is quantified in tons per hectares. Table 1 summarizes descriptive regional information, where we use the following acronyms for the regions: NMM for Northern Midland and Mountainous region (14 provinces), NCC for North Central Coast region (13 provinces), CHR for Central Highlands region (5 provinces), SR for Southeast region

¹IRRI is an organisation that promotes research and development of rice production in the world. Information about the institute can be found at <https://www.irri.org/>

(6 provinces), MDR for Mekong Delta region (13 provinces) and RRD for Red River Delta region (12 provinces). This table shows that rice yield is the highest in the RRD region. Figure 3 provides an overview of the overall evolution of rice yield over the considered period for the six agronomic regions where annual rice yield has been interpolated to improve the visualization of the trend. After a period of stagnation between 1987 and 1992, rice yield has exhibited consistent growth since 1992, affecting all Vietnamese provinces. This growth may be attributed to the progress of agronomic techniques over the years. While we lack a direct proxy for this progress, we will account for it through the incorporation of a linear time trend. This choice is supported by Figure 3.

Fig. 3 Average rice yield by agronomic regions from 1987 to 2016 (Units: Tons per hectares)

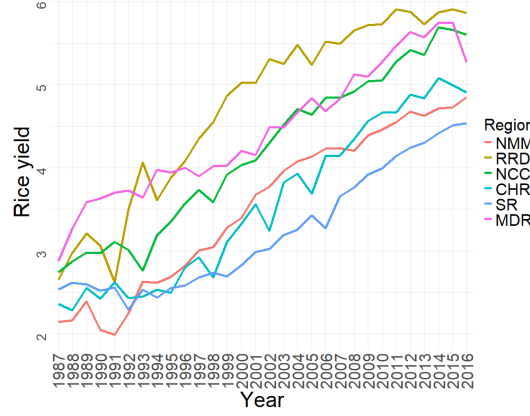


Table 1 Descriptive statistics by regions: Mean and (standard deviation)

Variables	NMM	RRD	NCC	CHR	SR	MDR
Number of observations	390	390	390	150	180	390
Total precipitation (Thousand ml per year)	1.47 (0.24)	1.40 (0.26)	1.30 (0.55)	1.67 (0.29)	1.24 (0.76)	1.31 (0.36)
Rice yield (Tons per hectare)	3.52 (1.12)	4.73 (1.24)	4.19 (1.11)	3.55 (1.14)	3.23 (0.92)	4.45 (0.98)

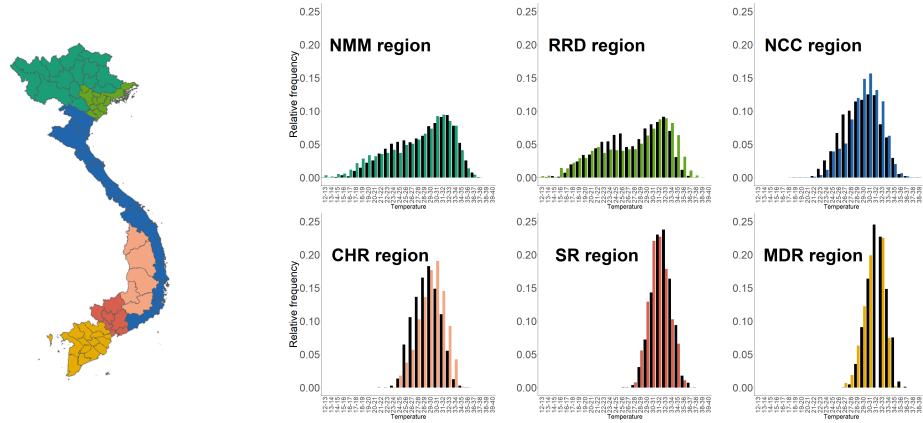
3.2 Weather data and its time evolution

The weather data used in this study encompasses daily maximum and daily minimum temperatures as well as total yearly precipitation records (see Table 1 for its descriptive analysis by region). Temperature data comes from the Climate Prediction Center (CPC) database developed and maintained by the National Oceanic and Atmospheric Administration (NOAA). [17] retrieved historical information pertaining to daily maximum and minimum temperatures for a grid with a resolution of 0.50×0.50 degrees

of latitude and longitude, specifically for the geographical expanse of Vietnam. Subsequently, they have transformed this data to yield the daily maximum and minimum temperature for each of 63 Vietnamese provinces and during a period of 30 years (1987-2016) (365 or 366 values for each year). To summarize, the original observation unit in our study is indexed by year and province and for each unit we therefore have an annual scalar rice yield value and a daily evolution of maximum and of minimum temperature. After the smoothing, the compilation yields two temperature distributions (minimum and maximum) for each of the 1890 statistical units.

Figure 4 displays the Vietnamese map on the left hand side colored by region. On the right hand side, for each of the six regions, we show the smoothed densities averaged over the first three and last three years of our study period, with the same color scheme, where average is understood with the Bayes space operations as defined in Section 2.2. The right part of the figure emphasizes the regional disparities: the northern regions have a sub-tropical climate with four seasons whereas the southern regions have a tropical climate with two seasons. Concerning the historical climate change, we can see that the Mekong Delta region is the most affected by climate change with a higher concentration of temperatures around $28^{\circ}C$ and $32^{\circ}C$ for maximum temperature ($17^{\circ}C$ to $29^{\circ}C$ for minimum temperature, see supplementary material). Similarly in the RRD region, the temperature range most affected by the change is within $27^{\circ}C$ to $33^{\circ}C$.

Fig. 4 Map of Vietnam and regional average histogram of maximum temperature during the first three years 1987-1988-1989 (black) and last three years 2014-2015-2016 (color as region in the map)



To enable integration of the smoothed histograms into the regression model, it is essential to represent them using a common CB-spline basis. Consequently, we must employ a consistent set of knots across all $63 * 30 = 1890$ histograms. For this reason, we first pool all observations into a single distribution and place the knots at the quantiles of this global distribution.

Improving this phase of the process hinges on obtaining information about the specific starting and ending dates of the growing season within each province. However since these temporal boundaries may exhibit substantial variability across geographical

regions as we have seen, the adoption of a standardized temperature range across all provinces would then be rendered difficult, unless we find a way of overcoming this technical constraint.

As for future changes, the Intergovernmental Panel on Climate Change (IPCC) provides projections of global CO_2 emissions and associated temperature distributions around the world under several scenarios associated to representative concentration pathways (RCPs) for the end of this century, see for example [18]. In Section 5, we will use the most optimistic RCP called RCP2.6, which projects an average increase of $1^\circ C$ relative to the period 1986-2005. The RCP2.6 data for Vietnamese provinces come from [19].

4 The scalar-on-composition and scalar-on-density regression models

Using regression models to relate agricultural yield and climate descriptors is by no means a new endeavor, as evidenced by [20] and [2]. Climate change exerts both direct and indirect impacts on various facets of the food system encompassing food production, storage, processing, distribution, retail and consumption, as discussed by [21]. Due to its direct exposition to weather conditions, crop production is all the more sensitive to climate change (see for example [22]).

Within the field of econometrics, assessing the impact of climate change for a given economic sector relies on the specification of the relationship between a change in the climate indicators and the corresponding change in the outcome. Noteworthy, among the contributions related to crop yield and climate are the insights provided by [3] and [23], while a recent and comprehensive survey can be found in [24]. [3] build their assumptions on the premise that temperature effects on yields accumulate over time and that yield is proportional to total exposure. In other words, the order in time in which the temperatures occur is considered to have no impact on the yield. This assumption justifies treating temperature density as a functional covariate rather than temperature as a function of time, as in [25]. In [3], the regression model incorporates the temperature density via an integral against a functional regression coefficient. This regression parameter, itself a function of temperature, encapsulates the sensibility of crop yield at different temperature levels. The implementation of the functional model of [3] uses an approximation of the integral on the right hand side. After approximation, the covariates are the histogram bin relative frequencies, with one bin omitted to account for the fact that the sum of these relative frequencies is 1. This strategy has been adopted by several researchers, gaining prominence after its use in [26]. For instance, [27] applied this approach in their study of how subsistence Peruvian farmers respond to extreme heat.

While modeling the linear effects of temperature bin frequencies as in [3] aligns with absolute differences in histogram objects, compositional data analysis, in contrast, is consistent with relative differences. Therefore the compositional model is compatible with relative changes of the densities, which is more coherent for these objects. Our objective is to showcase a regression model to unravel the relationship between rice yield and the distribution of daily maximum and minimum temperatures for the

corresponding year and province, while also controlling for additional covariates, and respecting the nature of the distributional temperature data by using log-ratio of frequencies as covariates. Unlike a conventional time series model used for yield prediction in the future, our focus here is to leverage spatio-temporal variability to quantify the influence of temperature on rice yield. Since the focus is more on demonstrating a proof of concept than conducting a full case study, we keep the model as simple as possible. This is why we restrict attention to linear (in the simplex sense) functionals. Similarly, we decide to include a simple linear time trend in the model as a proxy for unobserved factors that may have evolved over time, such as advancements in production techniques. In view of Figures 3, the inclusion of a linear trend appears to be a reasonable choice. We further use other controlling factors namely precipitation and regional dummies.

The choice thus boils down to either utilizing a histogram of daily temperatures as a compositional covariate, akin to the approach in [5], or opting for a smoothed representation of the temperature density as a continuous density covariate, following the method outlined in [11]. Before delving into the results, let us first revisit the fundamental principles behind these two models.

4.1 The scalar-on-composition regression model

The scalar-on-composition regression model as presented for example in [5] constitutes a regression framework where at least one of the covariates takes the form of a compositional vector. In our discrete regression setup, the compositional vectors are vectors of relative frequencies of bins which can also be viewed as discrete densities. Any linear function of a compositional explanatory variable $\mathbf{X} \in \mathcal{S}^D$ must be of the form $\langle \beta, \mathbf{X} \rangle_A$, where β is a parameter vector of $\mathcal{S}^D$ and $\langle \cdot, \cdot \rangle_A$ is the classical Aitchison inner product in $\mathcal{S}^D$ (see e.g. [4]). Therefore a linear model designed to explain a scalar variable Y with possibly several compositional variables $\mathbf{X}_j \in \mathcal{S}^{L_j}$ for $j = 1, \dots, J$ and several scalar variables $\mathbf{Z}_l$ for $l = 1, \dots, L$ is formulated by an equation of the form

$$Y_i = \alpha + \sum_{j=1}^J \langle \beta_j, \mathbf{X}_{ij} \rangle_A + \sum_{l=1}^L \gamma_l \mathbf{Z}_{il} + \epsilon_i, \quad (3)$$

where the parameters γ_l of the scalar variables are scalar, the parameters β_j of the compositional variables belong to $\mathcal{S}^{L_j}$ and the errors ϵ_i are i.i.d. gaussian variables with mean zero and variance σ^2 . Note that this model is not linear in the classical sense with respect to the $\mathbf{X}_j$ variables, which is coherent with [3]. For our application, it is essential to index all observations according to both the province i and the year k therefore the single index i of equation (3) is from now on replaced by the two indices i and k . This adjustment allows us to define Y_{ik} as the rice yield for province i (ranging from 1 to 63) in year k (spanning from 1 to 30). Initially, the model comprises several classical scalar variables ($L = 7$) including time, precipitation and five regional dummies (reference region being CHR). This choice of covariates is pretty much the same as in [3] (precipitation, regional fixed effects and time trends). In addition to these, we also incorporate two discrete densities as compositional covariates, namely the histograms of maximum and minimum daily temperature, reported with equal bins

of length $1^\circ C$. Moreover, after testing the inclusion of interactions between the two discrete densities and the five regional dummies, we decide to integrate the interactions solely for maximum temperature and two specific regions: RRD, and NCC. As a result, we get $J = 4$ compositional parameters associated with the discrete densities and denoted by $\beta_{RRD}^{\max}$, $\beta_{NCC}^{\max}$ and $\beta_{other}^{\max}$ for the maximum temperature and $\beta^{\min}$ for minimum temperature.

As demonstrated for example in [28], after transformation of the compositional covariates by any transformation in the log-ratio family (isometric or additive log-ratio, see [4] for detail), the estimation of such a model can be done by ordinary least squares. The choice of any of these transformations correspond to a particular parametrization of the same model but yield the same result for the discrete densities contribution when expressed as a linear combination of the logarithm of the histogram bin frequencies (with a zero sum constraint on the coefficients). Although the clr transformation is not adapted to the model fitting step since it would lead to a singular design matrix for the model, it is useful to write the clr version of equation (3) for interpretation purposes:

$$Y_i = \alpha + \sum_{j=1}^J \langle \text{clr} \beta_j, \text{clr} \mathbf{X}_{ij} \rangle_{\mathbb{R}^{L_j}} + \sum_{l=1}^L \gamma_l \mathbf{Z}_{il} + \epsilon_i. \quad (4)$$

4.2 The scalar-on-density regression model

Extending the model in [11] to the case of several density covariates as well as additional scalar covariates, we consider the following linear scalar-on-density regression model

$$Y_i = \beta_0 + \sum_{j=1}^J \langle \beta_j(t), f_{ij}(t) \rangle_{\mathcal{B}^2([a_j, b_j])} + \sum_{l=1}^L \gamma_l \mathbf{Z}_{il} + \epsilon_i, \quad (5)$$

where Y_i is the scalar dependent variable, β_0 is a real intercept, Z_l ($l = 1, \dots, L$) are real-valued covariates with their corresponding parameters γ_l , and ϵ_i are normal errors with mean zero and standard deviation σ^2 . The curve densities $f_{ij}(t)$ as well as the corresponding curve-parameters $\beta_j(t)$ (referred to as beta curves later on) are functions of the temperature argument t and are assumed to belong to some Bayes space $\mathcal{B}^2([a_j, b_j])$. As for model (3), model (5) is non-linear in the classical sense with respect to the density covariates.

Using the fact that the clr transform is an isometry between $\mathcal{B}^2([a, b])$ and $L_0^2([a, b])$ equipped with their respective inner products, we can rewrite the model as follows

$$Y_i = \beta_0 + \sum_{j=1}^J \langle \text{clr} \beta_j(t), \text{clr} f_{ij}(t) \rangle_{L_0^2([a_j, b_j])} + \sum_{l=1}^L \gamma_l \mathbf{Z}_{il} + \epsilon_i. \quad (6)$$

In order to estimate model (6), we first need to use a basis expansion of the functional parameters $\beta_j(t)$, as well as a similar expansion for the densities $f_{ij}(t)$. For the sake of simplicity, we will use the same basis system to express the functional regression parameters and the observed functional explanatory variables. The expansion can be written directly in $\mathcal{B}^2([a, b])$ with a basis of CB-splines or equivalently

for the clr transforms in $L_0^2([a, b])$ with a basis of ZB-splines. With the notation $\text{clr}\beta_j(t) = \sum_{l=1}^{g+k} z_l(\beta_j)Z_l(t)$ and $\text{clr}f_j(t) = \sum_{l=1}^{g+k} z_l(f_j)Z_l(t)$, where $z_l(h)$ are the coordinates of $\text{clr}h(t)$ for a density h in the basis of ZB-splines Z_l , ($l = 1, \dots, g+k$), we replace these functions by their expansions in the inner products of the model equation (5) or (6). Consequently, the inner products terms appear as linear combinations of the beta curves coordinates whose coefficients are given by the product of the Gram matrix (inner products of all pairs of basis functions $\langle Z_l, Z_m \rangle_{L_0^2}$) by the densities coordinates as in [11] since for any j we have $\langle \sum_{l=1}^{g+k} z_l(\beta_j)Z_l(t), \sum_{m=1}^{g+k} z_m(f_j)Z_m(t) \rangle_{L_0^2} = \sum_{l=1}^{g+k} \sum_{m=1}^{g+k} z_l(\beta_j)z_m(f_j)\langle Z_l, Z_m \rangle_{L_0^2}$. After this step, we are back to a classical linear model for ordinary covariates that we can fit with ordinary least squares. As in [11], we use the low dimensional regression approach which keeps the dimensionality of the basis small relative to the sample size and thus does not require penalization.

In our application, we have the same $L = 7$ classical covariates for the scalar-on-composition and the scalar-on-density models (time, precipitation and regional dummies) with their corresponding parameters γ_l . In the scalar-on-composition model the maximum and minimum temperature distributions are represented in the model by two histogram covariates whereas in the scalar-on-density model, they enter as two smooth density covariates, denoted by $f_{ik}^{\max}$ and $f_{ik}^{\min}$ in province i and year k . To facilitate the comparison, we include the same interactions between densities and regional dummies. The corresponding curve-parameters will be denoted by $\beta_{RRD}^{\max}(t)$, $\beta_{NCC}^{\max}(t)$ and $\beta_{\text{other}}^{\max}(t)$ for the maximum temperature and $\beta^{\min}(t)$ for minimum temperature. Finally ϵ_{ik} are normal errors with mean zero and standard deviation σ^2 . $f_{ik}^{\max}$, $f_{ik}^{\min}$ as well as all the curve-parameters are assumed to belong to some Bayes space $\mathcal{B}^2([a, b])$ (a and b will differ for maximum and minimum temperature).

The number of basis functions for the expansion is a function of the number of knots. In order to reduce variability, it is advisable to use a small number of knots compared to the sample size. Respecting the Schoenberg-Whitney conditions of Section 2.4, we select $g = 9$ knots corresponding to the dimension $9 + 3 = 12$ for the corresponding ZB-basis.

A technical but important detail for comparing the two models is that for a bin of size 1 the discrete inner product of two histograms correspond exactly to their smooth inner product.

Let us note an important difference between the scalar-on-composition and scalar-on-density models. Conventional compositional data analysis does not pay attention to the order of the components (permutation invariance). However in our case, for a temperature histogram, the components correspond to temperature bins and the order of these bins should be considered in order to account for some continuity of the bin frequencies with respect to the bins centers on the temperature axis. In contrast, the functional density approach does take this into account.

4.3 Model results

We implemented the Schlenker and Roberts' model approach as well as the two models presented in Sections 4.1 and 4.2, and the annual mean temperature alternative (hereafter simple model). The histograms smoothing step and the fitting of both

compositional models are performed with the R packages *compositions* and *robCompositions*, adapting some codes from [29]. The comparison between Schlenker and Roberts’ model and the scalar-on-composition model shows very similar adjusted R squared (0.8 for the first versus 0.79 for the second). However, as we already mentioned, it is due to the methodological drawback that we discard the Schlenker and Roberts’s model. The plot of residuals as a function of fit do not reveal any heteroscedasticity (see supplementary material for diagnostics). The parameter estimates for classical significant variables displayed in Table 2 are comparable between scalar-on-composition and scalar-on-density models. Moreover, we performed some tests in the scalar-on-composition model showing that the maximum temperature and minimum temperature histograms are statistically very significant (with p-value less than 10^{-16}).

Table 2 Estimated coefficients associated to regional dummies, total precipitation and year

Variable	Regression type	
	scalar-on-composition	scalar-on-density
Constant	2.62*** (0.31)	2.72*** (0.26)
Region (Reference = CHR)		
NMM	0.06 (0.27)	−0.09 (0.22)
RRD	−0.22 (0.37)	−0.05 (0.30)
NCC	−0.75** (0.35)	−0.86*** (0.30)
SR	−1.18*** (0.15)	−1.25*** (0.13)
MDR	−0.09 (0.27)	−0.14 (0.14)
Total precipitation	−0.0002 (0.04)	0.01 (0.04)
Year	0.10*** (0.002)	0.10*** (0.002)
Adjusted R ²	0.79	0.78
Residual Std. Error	0.56 (df = 1780)	0.57 (df = 1834)
F Statistic	65.08*** (df = 109; 1780)	123.46*** (df = 55; 1834)
RMSE	1.10	0.56

Note: *, **, and *** mean significant at 10%, 5%, and 1%, respectively

Table 3 displays the mean, median and standard deviation of the squared distance between fitted and observed values for the four models and the six regions. The scalar-on-density model with 56 parameters has a better fit than the three other ones. The scalar-on-composition model and the Schlenker and Roberts’s model, both with 110 parameters, are comparable (with a small advantage for the scalar-on-composition) and not so different from the simple model with 12 parameters. For evaluating prediction performance, we perform a leave one-year-out cross-validation and compute the corresponding prediction errors. The results in Table 4 show that the scalar-on-density model performs best again.

4.4 Parameters interpretation

The parameters relative to temperature are depicted on Figure 5 in clr-space for the scalar-on-composition model and on Figure 6 in Bayes space on the left and clr-space on the right for the scalar-on-density model. The figures will help us get an idea of the temperature ranges of high importance for rice yield and more quantitative statements will follow from equations of [30]. Looking at the model equations in clr space, we

Table 3 Mean squared distance between observed and fitted values per province by models and regions (median, mean and standard deviation)

Regression model						
	NMM			RRD		
	Median	Mean	Sd	Median	Mean	Sd
Annual mean	0.45	1.46	2.74	0.70	1.35	1.64
Schlenker and Roberts	0.53	1.46	2.58	0.73	1.38	1.74
Scalar-on-composition	0.51	1.48	2.63	0.74	1.42	1.80
Scalar-on-density	0.12	0.23	0.30	0.18	0.40	0.58
	NCC			CHR		
	Median	Mean	Sd	Median	Mean	Sd
Annual mean	0.27	0.61	0.82	0.52	1.49	2.30
Schlenker and Roberts	0.32	0.69	0.90	0.49	1.50	2.26
Scalar-on-composition	0.32	0.70	0.94	0.63	1.53	2.38
Scalar-on-density	0.08	0.22	0.33	0.14	0.31	0.39
	SR			MDR		
	Median	Mean	Sd	Median	Mean	Sd
Annual mean	0.66	1.25	1.46	0.43	0.99	1.50
Schlenker and Roberts	0.91	1.42	1.59	0.46	1.02	1.54
Scalar-on-composition	0.79	1.34	1.55	0.44	1.05	1.61
Scalar-on-density	0.24	0.36	0.42	0.14	0.40	0.90

Table 4 Leave-one-year-out cross-validated mean squared prediction error per province (median, mean and standard deviation)

Regression model	Median	Mean	Sd
Annual mean	0.33	0.37	0.10
Schlenker and Roberts	0.33	0.38	0.14
Scalar-on-composition	0.32	0.36	0.12
Scalar-on-density	0.31	0.35	0.12

can guess that temperature intervals of maximum impact on yield will correspond to intervals of oscillations from positive to negative values of the clr parameters and this is formally confirmed by [30] as we explain below. Observe that the oscillations of the smooth clr curves on the right of Figure 6 follow the same pattern as in the discrete case. These effects are also evident in the corresponding smooth beta curves on the left, where the temperature intervals of high importance for rice yield are clearly visible across the three regions.

Quantitative covariates marginal impact in scalar-on-composition regression can be evaluated either using finite increments as in Coenders and Pawlowsky-Glahn [28] or infinitesimal increments as in Morais et al [31] but the two approaches coincide in this particular case as can be seen in [30].

The evaluation of marginal effects typically involves examining how changes in the covariate affect the predicted dependent variable. Adapting this approach to a compositional covariate, [30] propose to use changes of a compositional explanatory variable X (in our case a histogram X_{ij}) along a linear path parametrized by a scalar

Fig. 5 Estimated clr coefficients of maximum temperature for RRD region in the scalar-on-composition model

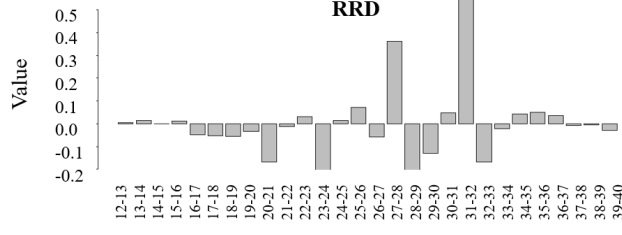
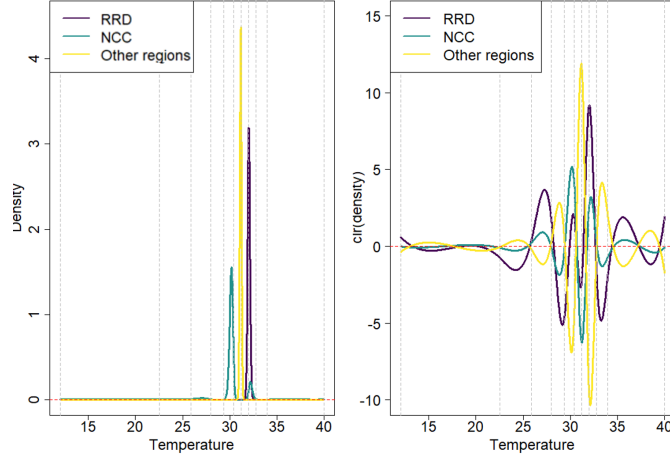


Fig. 6 Curves of $\hat{\beta}^{\max}$ in the scalar-on-density model with interactions for all regions. Left: in B^2 , right: in L^2

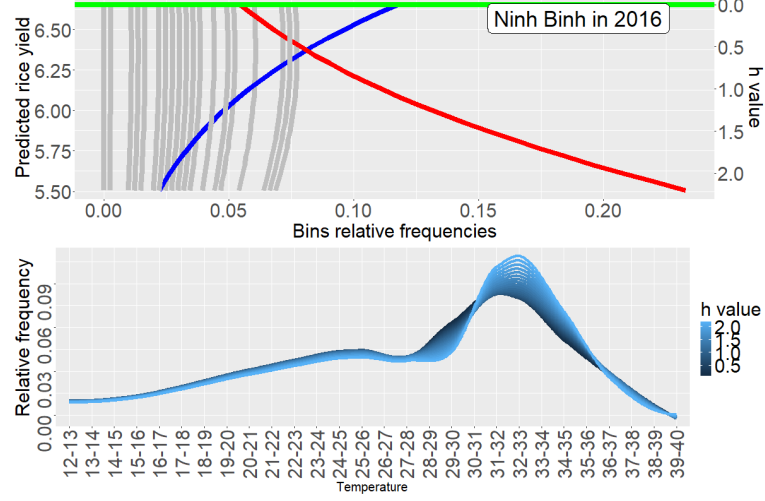


h and a direction φ in the variable's simplex as follows

$$X(h) = X(0) \oplus h \odot \varphi, \quad (7)$$

where $X(0)$ is an initial observed value of the compositional covariate (at an observed data point). Since many linear paths are conceivable, we focus on those that facilitate interpretation, in particular the relative compensation scheme, where one component of the covariate is relatively increased at the expense of another component while keeping all other pairwise ratios constant. Given that the order of the bins is relevant in our case, we propose examining such changes for consecutive components, for instance, increasing the density of bin $k + 1$ at the expense of the previous bin k . [30] describe the corresponding changes in the predicted dependent variable as a function of the components of the direction of change φ_k and the pairwise log-ratio of the parameter components. Let us first understand the impact of such an evolution on the example of Ninh Binh province (RRD region) in 2016. The top plot of Figure 7 illustrates the evolution according to equation (7) of the maximum temperature histogram relative

Fig. 7 Parameter interpretation: on top, evolution of rice yield on the vertical axis as a function of bin's shares on the horizontal axis along a relative compensation path starting at Ninh Binh province in 2016; on bottom, evolution of the temperature density along the same path as a function of h . The red curve is for the bin 32-33 share, the blue curve for the bin 31-32 and the grey curves for the other shares.



frequencies (on the horizontal axis) as the share of the 32–33 bin increases (red line) at the expense of the 31–32 bin (blue line), while the ratios involving the other shares (grey line) remain constant, while h increases from 0 to 2.2. Although these other shares adjust due to the compositional constraint, they remain relatively stable. The vertical axis of the top plot represents the corresponding change in predicted rice yield $\hat{Y}(h)$ while other covariates remain at their Ninh Binh 2016 value. As h increases from the initial value of 0, rice yield on the vertical axis decreases from its initial position of 6.65 in Ninh Binh province in 2016. The bottom plot of Figure 7 shows the corresponding evolution of the maximum temperature density. Note that despite the fact that we are dealing with the histogram covariates here, we have performed a smoothing for a better visualization of the density change. We can see that the number of days with maximum daily temperature higher than 31°C gets larger at the expense of the number of days when this temperature is lower. The model predicts a decline of rice yield when the density change moves in that particular direction.

For a more quantitative assessment, using [30] the infinitesimal marginal effect of the evolution considered in Figure 7 is given by $\frac{1}{2} \log \frac{\beta_{[a_{k+1}, a_{k+2}]}}{\beta_{[a_k, a_{k+1}]}}$ where the k^{th} bin is denoted by $[a_k, a_{k+1}]$. The variance of this value is easily computed (see appendix). The significant pairs of bins for the RRD region at level 5% are the following: 26-27 versus 27-28, 27-28 versus 28-29, 31-32 versus 32-33 and 36-37 versus 37-38. Taking for example the contrast 31-32 versus 32-33, the infinitesimal marginal effect of the change φ_{31} is equal to $\frac{1}{2} \log \frac{\beta_{32-33}}{\beta_{31-32}} = -0.37$ which is negative from Figure 5, as the difference in logarithms is equal to the difference in clr values. This means that a relative increase of the relative frequency of bin 32-33 versus bin 31-32 is associated

to an additive change in predicted rice equal to $-0.37h$. Figure 5 also shows that this contrast is the largest in the RRD region. For the NCC region, none of the contrast are significant. For the Other region, we get three significant contrasts: 14-15 versus 15-16, 30-31 versus 31-32 and 31-32 versus 32-33. For minimum temperature, the significant contrasts are 14-15 versus 15-16 and 25-26 versus 26-27 (see supplementary material).

Similarly in the smooth case, and using on purpose the same notation for a different object, we can consider finite linear paths in the Bayes space $\mathcal{B}^2([a, b])$ such as (7) where φ now denotes a direction curve in $\mathcal{B}^2([a, b])$ and the histogram X is replaced by a density f , and thus obtain a figure similar to 7. However infinitesimal marginal impact interpretations akin to those in [30] have yet to be developed.

The comparison of the smooth beta curves with their discrete histogram counterparts seem to imply higher maximum values for the smooth curves. This is the same phenomenon as when the smooth density estimate is larger than the histogram counterpart in regions of high density of observations.

The detection of the influential intervals may be affected by the choice of parameters: in the discrete case by the bin size choice and the minimum of the first bin and in the smooth case by the number of knots of the spline approximation. However it is known that a small change in the end-point of the first bin can dramatically affect the histogram whereas the smooth approach is less sensitive to the number of knots. Overall we can say that the results of both models are coherent but more precise and possibly less sensitive to parameter choices for the scalar-on-density model. Moreover unlike the scalar-on-composition model, the scalar-on-density model takes advantage of the information about the order of observations on the temperature axis.

5 Climate change scenario and its marginal effect

Given the conclusions of the previous section, we focus on the scalar-on-density model. Technically the computations are similar but easier in the scalar-on-composition case and the interested reader can find details in the supplementary material.

We would like to compute projections for the marginal impact of temperature on rice yield corresponding to the IPPC projections of the temperature at the end of this century, see for example [32], and more precisely those projected by RCP2.6.

The negative perturbation in the Bayes space is defined by

$$X_1 = X_0 \oplus \varphi \Leftrightarrow \varphi = X_1 \ominus X_0. \quad (8)$$

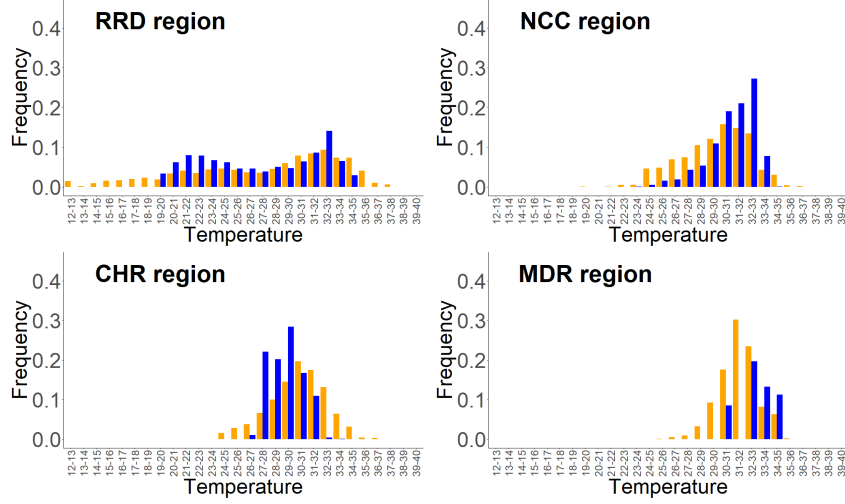
Let $s_0 = 2016$ be the last observed time and $s_1 = 2099$ the end of the century. For a province i , let us define the direction of change $\varphi_i \in \mathcal{B}^2(a, b)$ (between s_0 and s_1) by the negative perturbation of X_{is_1} by X_{is_0} , so that

$$\varphi_i = X_{is_1} \ominus X_{is_0} = \mathcal{C}\left(\frac{X_{is_1}(t)}{X_{is_0}(t)}\right). \quad (9)$$

Let $\varphi_i^{\max}$ and $\varphi_i^{\min}$ denote the change curves for maximum temperature and minimum temperature between the last observed date 2016 and the end of the century

2099 given by RCP2.6. In order to first visualize these changes, we plot some histograms of the RCP2.6 compared to the corresponding 2016 histogram on Figure 8 for maximum temperature. The four subplots correspond to the four selected regions and the regional change has been computed as a simplex average of the province changes in that region. We observe that for the RRD and NCC regions, it is the temperature bin 32-33 which is experiencing the highest change, whereas it is the bin 31-32 in the MDR region and the bin 29-30 in the CHR region.

Fig. 8 Maximum temperature distributions in four regions: observed in 2016 (in orange) and projected by RCP2.6 (in blue)



If this hypothetical climate change were to happen in a given province, we could now compute a projection of the resulting rice yield change, breaking it down into the contribution from the change of the maximum temperature distribution and that of the minimum temperature. Let $\hat{Y}_{is_1}(\varphi_i^{\max}) - \hat{Y}_{is_0}$, respectively $\hat{Y}_{is_1}(\varphi_i^{\min}) - \hat{Y}_{is_0}$, denote the projected rice yield change due to maximum temperature, respectively minimum temperature, under the change scenario. Given that our model is linear for the Bayes structure, the resulting change of rice yield due to the change of maximum temperature for a given province i in the scalar-on-density regression model is given by

$$\begin{aligned}\hat{Y}_{it_1}(\varphi_i^{\max}) - \hat{Y}_{it_0} &= \sum_{r=1}^3 \mathbf{1}_{i \in r} \langle \hat{\beta}_r^{\max}(t), \varphi_i^{\max}(t) \rangle_{\mathcal{B}^2} \\ &= \sum_{r=1}^3 \mathbf{1}_{i \in r} \langle \text{clr} \hat{\beta}_r^{\max}(t), \text{clr} \varphi_i^{\max}(t) \rangle_{L_0^2}.\end{aligned}$$

Similar formulas can be written for minimum temperature impacts. Note that since a given province belongs to a single region, there is indeed a single non-zero term

in the right hand side sums. The inner products $\langle \text{clr}\hat{\beta}_r^{\max}(t), \text{clr}\varphi_i^{\max}(t) \rangle_{L_0^2}$ and $\langle \text{clr}\hat{\beta}_r^{\min}(t), \text{clr}\varphi_i^{\min}(t) \rangle_{L_0^2}$ therefore characterize the impacts of a change in temperature density in the scalar-on-density model in region r . For simplifying their computation, we have approximated the smooth curve in the denominator of (9) $X_{is_0}(t)$ by the original histogram X_{is_0} since the numerator is only available as histogram data. The computation of the variance of the impacts is derived in the Appendix. They involve the computation of integrals of the products between the change functions and the ZB-spline basis functions. We show that these can be easily computed using the fact that our change functions are approximated by step functions combined with the fact that ZB-splines of order d are derivatives of B-splines of order $d + 1$, see [13].

Table 5 displays the impacts and their standard error in our application, together with a 95% confidence interval. The scalar-on-density model predicts a positive and significant impact of maximum temperature in all regions. The impact of minimum temperature is significant for all regions. It is positive except for the CHR and RRD regions where it partially compensates a large positive impact of maximum temperature. Even though the predicted rice yield change due to maximum temperature is negative in Section 4.4 for Ninh Binh and positive in Section 5 for all regions, the results are not contradictory. The difference reflects the fact that the direction of change predicted by the IPCC scenario differs from a simple relative compensation scheme which is a pedagogical tool to understand local effects on the temperature range.

Table 5 Impact of temperature change on rice yield by regions based on RCP2.6 for scalar-on-density regression (Units: Tons per hectares)

Region	Tmax				Tmin			
	mean	lwr	upr	sd	mean	lwr	upr	sd
NMM	0.42	0.34	0.50	0.14	0.42	0.40	0.44	0.04
RRD	10.99	10.93	11.04	0.07	-1.29	-1.30	-1.28	0.01
NCC	1.61	1.55	1.68	0.12	1.47	1.46	1.47	0.01
CHR	22.72	22.55	22.88	0.31	-3.70	-3.71	-3.70	0.01
SR	8.53	8.51	8.54	0.02	0.52	0.51	0.53	0.01
MDR	8.52	8.50	8.54	0.04	0.01	0.01	0.02	0.01

A careful comparison (see supplementary material) with impacts from the scalar-on-composition model illustrates how using a histogram rather than a smooth curve can reduce precision. In the RRD region for example, the scalar-on-composition model is blind to the sign change of the clr occurring in the one degree range 32-33. The scalar-on-composition model results in smaller impacts: indeed the scalar-on-composition model inner products can be viewed as approximations of the smooth inner products obtained by replacing the integral of the product of clr functions over a given bin by the mean value of the product of clr functions multiplied by the bin length (one here). Therefore, the discrete product of clr values can be seen as approximating the mean values of the corresponding smooth product of clr functions within the bins, resulting in a reduction in size.

6 Conclusion

We have proposed and illustrated a procedure for assessing the impact of climate change on rice yield production in Vietnam using scalar-on-distribution regression with a compositional and a functional frameworks. These models account for the distributional nature of temperature covariates and may be viewed as both an extension and an improvement of the model proposed by [3]. They allow to detect the temperature intervals which are most influential for rice yield. We have illustrated the parameters interpretation for the scalar-on-composition model. More theoretical work needs to be done to interpret the curves in the scalar-on-density model. We have derived formulas to evaluate the variances of the semi-elasticities and of the marginal impact estimates. The functional approach allows to keep more information from the density objects: notably, the order of observations on the temperature axis, and consequently the smoothness of the curve's shape are ignored by the scalar-on-composition model. Moreover the functional approach is less sensitive to parameter choice and via the knots position allows to use a more complete information in the temperature ranges where the data points concentrate.

We consider our illustration as a feasibility demonstration which would need refinements in order to derive pragmatic projections. A more realistic assessment would require accounting for the cropping season in each region if the cropping season data were available. We did not have these data at hand and moreover dealing with density covariates with varying supports would have resulted in a higher complexity from the methodological side with different spline basis for each region. Both models are based on the assumption of [3] which could clearly be discussed. Indeed, effects might well accumulate while accumulating with different weight in different time spans. Provided one has the information about these time spans and a way to proxy the corresponding weights, these could be incorporated in our model using a weighted inner product, as in [33] and [34]. The spatial dimension of this data could also be incorporated in these models. With regard to long-term projections, note that errors arising from spline smoothing of input densities and even more importantly uncertainties in climate-model ensemble projections such as RCP2.6 are not captured in our current framework and may dominate the overall prediction error. A precise assessment of this effect would be a valuable objective for a future interdisciplinary project.

Alternative models dropping the linearity (in the simplex or Bayes sense) assumption while still respecting the distributional nature are possible using nonparametric functions of the inner-products as covariates. Another refinement of our models would be to consider Bayes spaces with a different support for each of the three considered subregions. In order to measure the impact of climate change, we have chosen to consider separately daily maximum and daily minimum temperature. It could be very interesting to use the bivariate density of these two characteristics, thus taking into account their probable correlation. This would require using bivariate constrained splines as in [35].

7 Appendix

7.1 Variance of the semi-elasticities

The semi-elasticity relative to the contrast between bin k , with bounds $[a_k, a_{k+1}]$, and bin $k + 1$ for $k = 1, \dots, 27$ is given by $\tau_k = \frac{1}{2} \log\left(\frac{\beta_{k+1}}{\beta_k}\right) = \frac{1}{2}(\text{clr}(\beta_{k+1}) - \text{clr}(\beta_k))$. It is then easy to derive its variance as follows from the variance-covariance matrix of the clr coefficients:

$$\text{Var}(\tau_k) = \frac{1}{4} \left[\text{Var}(\text{clr}(\beta_{k+1})) + \text{Var}(\text{clr}(\beta_k)) - 2\text{Cov}(\text{clr}(\beta_k), \text{clr}(\beta_{k+1})) \right]. \quad (10)$$

7.2 Variance of the impacts

This appendix provides details about the computation of the impacts and their variance. We derive the impact variance in a general scenario where the change direction curve may depend on province i and will use it for the change scenario given by the simplex-difference between the RCP2.6 histogram and the observed histogram in 2016 for that province. The computation is very similar in spirit for both the compositional and the functional framework, however the evaluation of the inner products involved is more intricate in the functional framework. We evaluate separately the impact of maximum temperature and that of minimum temperature, and then add them up to get the impact of climate change. We develop the computation for maximum temperature and the result for minimum temperature is obtained in the same fashion.

In the smooth framework, the impact estimate for maximum temperature between an initial time, say t_0 say 2016 and time t_1 say 2099, is given, for a province i in region r , by

$$\hat{Y}_{it_1} - \hat{Y}_{it_0} = \langle \text{clr}(\hat{\beta}_r^{\max}), \text{clr}(\varphi_i) \rangle_{L_0^2(a,b)}. \quad (11)$$

Because the RCP scenarios are available as histograms, we will assume that φ_i is a step function (constant on each bin with values $(\varphi_i)_j$ for bin j). Since $\text{clr}(\hat{\beta}_r^{\max})(t) = \sum_{l=1}^{g+3} z_l(\hat{\beta}_r^{\max}) Z_l^4(t)$ the impact of maximum temperature for province i in region r is then given by

$$\hat{Y}_{it_1} - \hat{Y}_{it_0} = \sum_{l=1}^{g+3} z_l(\hat{\beta}_r^{\max}) \int \text{clr}(\varphi_i)(t) Z_l^4(t) dt, \quad (12)$$

where $z(\hat{\beta}_r^{\max})$ is the $g + 3$ vector of components of the $\hat{\beta}_r^{\max}$ curve in the ZB-spline basis and $Z_l^4(t)$ is the l^{th} ZB-spline curve. For $i = 1$ to 63 and $l = 1$ to $g + 3$, let us denote by p_{il} the integral term

$$p_{il} = \int \text{clr}(\varphi_i)(t) Z_l^4(t) dt \quad (13)$$

and therefore $\hat{Y}_{it_1} - \hat{Y}_{it_0} = \sum_{l=1}^{g+3} z_l(\hat{\beta}_r) p_{il}$.

To compute the p_{il} , we take advantage of the fact that φ_i are constant on the bins (b_j, b_{j+1}) and then of the fact that the integral of a ZB-spline can be obtained with

differences of B-splines of a higher order using equation (7) in [13] as follows

$$p_{il} = \int_{12}^{40} \text{clr}(\varphi_i)(t) Z_l^4(t) dt = \sum_{j=1}^{28} \int_{b_j}^{b_{j+1}} \text{clr}(\varphi_i)_j(t) Z_l^4(t) dt \quad (14)$$

$$= \sum_{j=1}^{28} \text{clr}(\varphi_i)_j \int_{b_j}^{b_{j+1}} Z_l^4(t) dt = \sum_{j=1}^{28} \text{clr}(\varphi_i)_j (B_l^5(b_{j+1}) - B_l^5(b_j)) \quad (15)$$

Turning now attention to the variance of the estimated impact of maximum temperature, the unbiasedness of the OLS estimates in clr space implies that $\mathbb{E}(z(\hat{\beta}_r^{\max})) = z(\beta_r^{\max})$. Therefore we have

$$\text{Var}(\hat{Y}_{it_1} - \hat{Y}_{it_0}) = \mathbb{E} \left[\left(\sum_{l=1}^{g+3} (z_l(\hat{\beta}_r^{\max}) - z_l(\beta_r^{\max})) p_{il} \right)^2 \right] \quad (16)$$

Let P be the $63 \times (g+3)$ matrix of elements p_{il} . Then the variance of the impact in province i is given by

$$\text{Var}(\hat{Y}_{it_1} - \hat{Y}_{it_0}) = \text{Var} P_i z(\hat{\beta}_r^{\max}) = P_i \text{Var}(z(\hat{\beta}_r^{\max})) P_i^T, \quad (17)$$

where P_i is the i^{th} row of P . We can estimate $\text{Var}(z(\hat{\beta}_r^{\max}))$ by the empirical variance-covariance matrix of the parameter estimates.

Because the computation of the variance would be lengthy and difficult to interpret, we decide to approximate it by using a single scenario of change by region thus replacing in (11) the change φ_i by the average (in the simplex sense) of the change corresponding to the RCP scenarios of all the provinces in that region. Therefore p_{il} is replaced by p_{rl} when province i is in region r and the matrix P is then $4 \times (g+3)$ resulting in four values for the right hand side of (17).

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